TAMILARASAN VELMURUGAN

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PROFESSIONAL SUMMARY

Software Engineer with 11 years of software development experience, specializing in C/C++, embedded and real-time systems, Linux/RTOS, and semiconductor factory automation. Hands-on experience building SECS/GEM host automation capabilities for semiconductor equipment, along with strong telecom Layer-2 and device-management expertise. Experienced in multithreaded software, debugging, feature ownership, customer issue resolution, and cross-functional product development from requirements through deployment and sustainment.

CORE TECHNICAL SKILLS

C/C++: C, C++11, object-oriented design, multithreaded/concurrent software, production debugging
Semiconductor Automation: SECS/GEM, host automation, equipment-to-host integration, FAB/MES workflows
Embedded & Real-Time: Embedded Linux, RTOS, VxWorks, Layer-2 (MAC), PTM/ATM, TCP/IP
Protocols & Management: SNMP, CLI, Netconf/YANG, MIB
Debugging & Quality: GDB, Valgrind, core-dump analysis, Google Test, Robot Framework, unit/integration/system testing
Tools & SDLC: Git, Mercurial, Jira, Bamboo, Visual Studio, Agile/Waterfall
Additional Programming: Python

PROFESSIONAL EXPERIENCE

Senior Software Engineer-II | Kulicke & Soffa — Singapore

May 2022 - Present

- Develop and sustain C/C++ software features for semiconductor equipment with demanding uptime and quality requirements.
- Develop Host Automation capabilities using SECS/GEM for equipment control and integration with semiconductor FAB/MES workflows.
- Own features end-to-end across requirements, design, implementation, integration, debugging, validation, deployment, and sustainment.
- Collaborate with Mechanical, System, Electrical, Vision, and Process teams to finalize system requirements and integration designs.
- Debug complex production and customer issues and deliver robust software changes across equipment and host automation interfaces.

Product Technical Lead | EdgeVerve (Infosys) — Bangalore, India

Dec 2020 - Nov 2021

- Developed HDFC Bank Wealth Management bond-report features with selectable PDF/XLS output using C++11, SQL, HTML, and JavaScript.
- Enhanced Edelweiss bond-creation workflows and simplified backend business logic in C++11 and SQL.
- Led engineers, conducted code reviews, and provided product and domain enablement to support high-quality delivery.

Technical Lead (Developer) | Nokia — Chennai, India

Jun 2015 - Oct 2020

- Designed and implemented Layer-2 auto-switching between VDSL2 (PTM) and ADSLx (ATM) with minimal service interruption and synchronized configuration management.
- Developed CPE software-upgrade interfaces aligned with ITU-T G.9701, integrating APIs with a Broadcom proxy and Google Protocol Buffers between application and hardware abstraction layers.
- Delivered E1 transport over SHDSL with CLI/SNMP interfaces, NT/LT inter-process messaging, OctalFALC framer integration, and Google Test unit coverage.
- Implemented Single Ended Line Testing (SELT) for NBN Co Australia using Netconf/YANG within the MOSWA architecture.
- Led offshore deliverables, HLD/design activities, code reviews, and technical training across MOSWA, Netconf/YANG, and RTOS.

EDUCATION

Bachelor of Technology, Computer Science Engineering | NIT Trichy | 2011 - 2015

TECHNICAL KEYWORDS

C++11 • C • Python • SECS/GEM • Embedded Linux • RTOS • VxWorks • Multithreading • OOP • Layer-2 MAC • PTM/ATM • TCP/IP • SNMP • Netconf/YANG • MIB • GDB • Valgrind • Git • Jira • Google Test • Robot Framework